

BEDFORD, United KingdomWMO: **035600**Lat: **52.2269N**Lon: **0.4653W**Elev: **84**StdP: **100.32**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) -2.4	(d) -5.6	(e) 2.4	(f) -1.2	(g) -4.2	(h) 2.7	(i) -0.4	(j) 13.8	(k) 8.9	(l) 12.5	(m) 8.8	(n) 3.3	(o) 60	(p) 0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.2	18.5	24.8	17.7	23.0	16.9	19.6	24.9	18.5	23.2	17.6	21.8	4.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	21.2	16.8	12.1	20.2	15.9	11.4	19.3	56.0	25.0	52.7	23.4	49.9	21.9	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.1	9.1	-6.3	30.4	2.5	1.9	-8.1	31.7	-9.5	32.9	-10.9	33.9	-12.7	35.3
			-6.5	21.7	2.4	1.9	-8.2	23.1	-9.6	24.2	-11.0	25.2	-12.7	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	4.5	4.8	6.5	8.8	11.7	14.8	17.1	17.0	14.5	11.0	7.2	4.8
	DBStd	5.45	3.33	3.32	3.03	2.98	2.92	2.62	2.53	2.53	2.47	2.86	3.11	3.70
	HDD10.0	781	172	146	114	56	15	1	0	0	1	21	92	163
	HDD18.3	2985	429	378	368	284	204	111	57	58	118	226	333	418
	CDD10.0	878	1	2	4	21	69	144	219	217	135	53	9	3
	CDD18.3	42	0	0	0	0	0	4	17	17	3	1	0	0
	CDH23.3	373	0	0	0	2	5	52	146	143	23	2	0	0
	CDH26.7	78	0	0	0	0	0	8	31	36	3	0	0	0
(14) Wind	WSAvg	4.5	5.2	5.2	5.0	4.5	4.4	4.0	3.9	4.0	4.2	4.6	4.6	4.9
(15) Precipitation	PrecAvg	600	52	39	33	46	45	49	52	56	54	60	61	51
	PrecMax	936	108	79	85	133	120	112	106	133	150	131	132	95
	PrecMin	391	10	2	3	3	4	8	16	2	11	13	24	15
	PrecStd	124	23	21	20	32	27	32	25	31	36	29	25	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.3	16.3	21.6	23.9	27.9	29.9	30.1	26.3	20.8	15.6	13.8
		MCWB	11.3	11.5	11.3	14.5	16.7	19.5	19.6	19.5	17.3	16.3	14.0	12.0
	2%	DB	11.9	12.1	14.2	18.1	21.8	24.5	26.9	27.2	22.9	18.0	14.3	12.7
		MCWB	10.4	10.0	10.5	12.9	16.1	17.8	18.5	18.5	16.8	14.8	12.7	11.3
	5%	DB	10.7	10.8	12.7	16.0	19.5	22.3	24.6	24.3	20.8	16.5	13.1	11.6
		MCWB	9.2	8.9	9.8	11.6	14.7	16.6	17.7	17.5	15.7	14.0	11.6	10.3
	10%	DB	9.5	9.8	11.4	14.2	17.3	20.4	22.7	22.3	19.2	15.4	11.9	10.4
		MCWB	8.1	8.1	8.9	10.5	13.3	15.4	16.7	16.5	14.9	13.3	10.6	9.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.8	12.2	15.5	18.7	20.7	20.7	21.2	19.4	16.9	14.3	12.5
		MCDB	12.6	13.6	14.7	19.9	21.8	26.6	26.9	26.8	23.6	19.6	15.1	13.4
	2%	WB	10.6	10.3	11.2	13.5	16.7	18.5	19.4	19.2	17.8	15.6	13.1	11.5
		MCDB	11.7	11.7	13.4	17.3	20.5	22.9	24.8	24.7	21.3	17.3	14.1	12.6
	5%	WB	9.4	9.3	10.3	12.2	15.3	17.2	18.4	18.2	16.7	14.6	11.9	10.5
		MCDB	10.5	10.6	12.3	15.1	18.6	21.2	23.3	23.0	19.7	16.1	12.9	11.5
	10%	WB	8.3	8.4	9.3	11.0	14.0	16.1	17.4	17.3	15.7	13.7	10.8	9.3
		MCDB	9.4	9.7	11.2	13.8	16.8	19.6	21.8	21.2	18.2	15.2	11.8	10.4
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.9	7.3	8.8	8.9	9.1	9.4	9.3	8.5	7.0	5.7	5.0
		MCDBR	6.1	6.7	9.0	11.5	12.5	12.5	13.0	12.7	11.1	7.7	6.0	5.6
	5% WB	MCWBR	5.8	5.3	6.1	7.0	7.5	6.8	5.8	5.6	5.7	5.2	5.2	5.2
		MCWBR	6.1	6.5	8.4	10.5	10.9	11.1	11.8	11.0	9.5	7.0	6.0	5.6
(40) Clear-Sky Solar Irradiance	taub		0.322	0.339	0.372	0.396	0.401	0.402	0.400	0.389	0.370	0.352	0.326	0.312
		taud	2.371	2.388	2.287	2.231	2.255	2.274	2.289	2.342	2.401	2.436	2.418	2.352
	Ebn at Noon		672	774	812	836	851	851	845	835	810	748	664	612
		Edh at Noon	61	79	105	125	128	127	123	111	93	74	58	52
(44) All-Sky Solar Radiation	RadAvg		0.77	1.46	2.53	3.88	4.77	5.15	4.93	4.08	3.06	1.78	0.94	0.62
	RadStd		0.07	0.18	0.31	0.43	0.31	0.51	0.52	0.30	0.27	0.15	0.10	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (91 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page